

3.5 Cherry-Picking Merge Conflict Scenario

I began the assignment by creating a new directory and initializing a fresh Git repository. After running `git init`, I created an initial file called `file1.js` containing simple placeholder content and committed it. This gave me a clean starting point, and VisualGit showed a single initial commit. I then created a new branch named `feature` and added three commits to it. The first two added `featureA.js` and `featureB.js`, and the third added `featureC.js` while also modifying `file1.js`. In that last commit, I rewrote `file1.js` to include the original content plus an additional line specific to the feature branch. This intentional change set up the conditions for a merge conflict later.

After finishing the feature branch work, I switched back to my default branch, which was named `master` rather than `main`. On this branch, I edited `file1.js` in a different way, keeping the initial content but adding a separate line that conflicted with the feature branch's version. I committed this change, and at this point the two branches had diverged in a way that guaranteed a conflict. VisualGit showed the two separate histories clearly.

Next, I identified the commit on the feature branch that contained the modification to `file1.js` and attempted to cherry-pick it onto `master`. As expected, Git reported a merge conflict in `file1.js`. When I opened the file, I saw the standard conflict markers showing the `master` version on one side and the feature branch version on the other. I manually resolved the conflict by choosing a combined version of the file, then staged the file and continued the cherry-pick. Git accepted the resolution and created a new commit on `master` representing the cherry-picked change.

During this process, I made a couple of mistakes that appear in the VisualGit history. At one point, I accidentally staged the file before resolving the conflict, which caused Git to abort the

Instruction Set:

```
mkdir cherry-pick-demo
cd cherry-pick-demo
git init
```

```
echo "//Initial content" > file1.js
git add . > git commit -m "Initial commit"
```

```
git checkout -b feature
echo "//Feature A" > featureA.js
git add . > git commit -m "Add Feature A"
```

```
echo "//Feature B" > featureB.js
git add . > git commit -m "Add Feature B"
```

```
echo "//Feature C" > featureC.js
echo "//Initial content" > file1.js
echo "//Feature C edit on feature branch" >> file1.js
git add . > git commit -m "Add Feature C and modify file1.js"
```

```
git checkout main
<edit file1.js //Initial content //Main branch edit>
```

```
git add . > git commit -m "Main branch edit to file1.js"
```

```
git log feature --oneline
<find hash for "Add Feature C and modify file1.js">
```

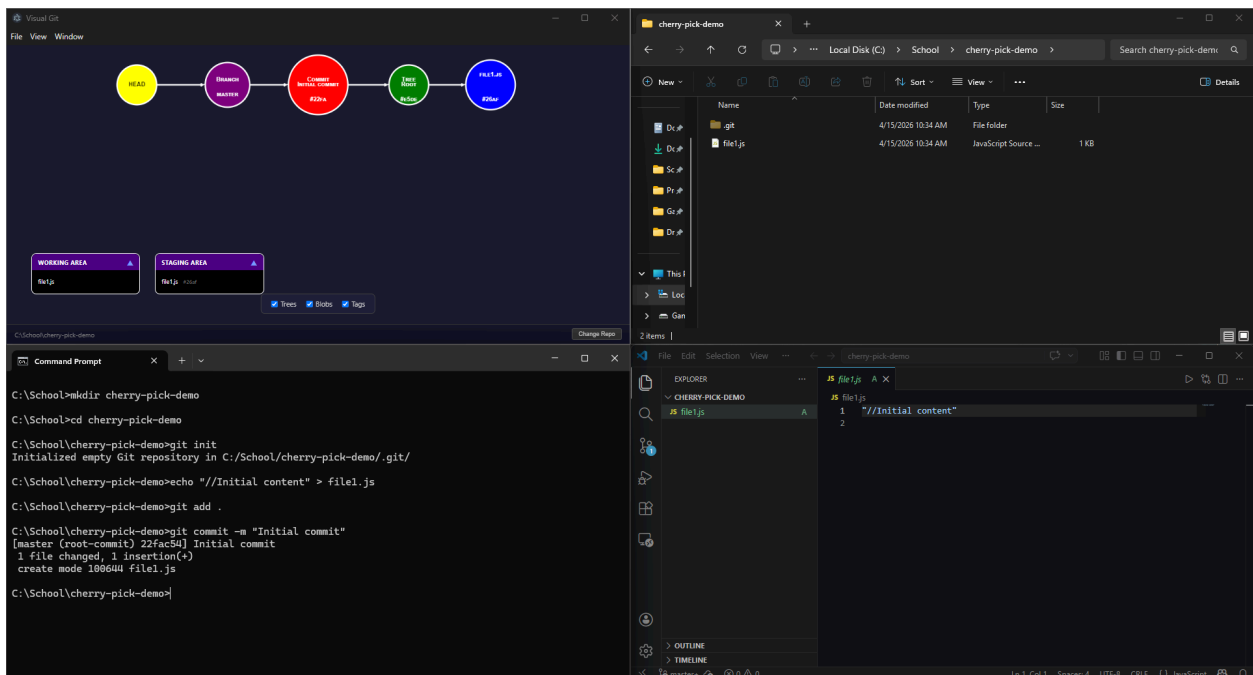
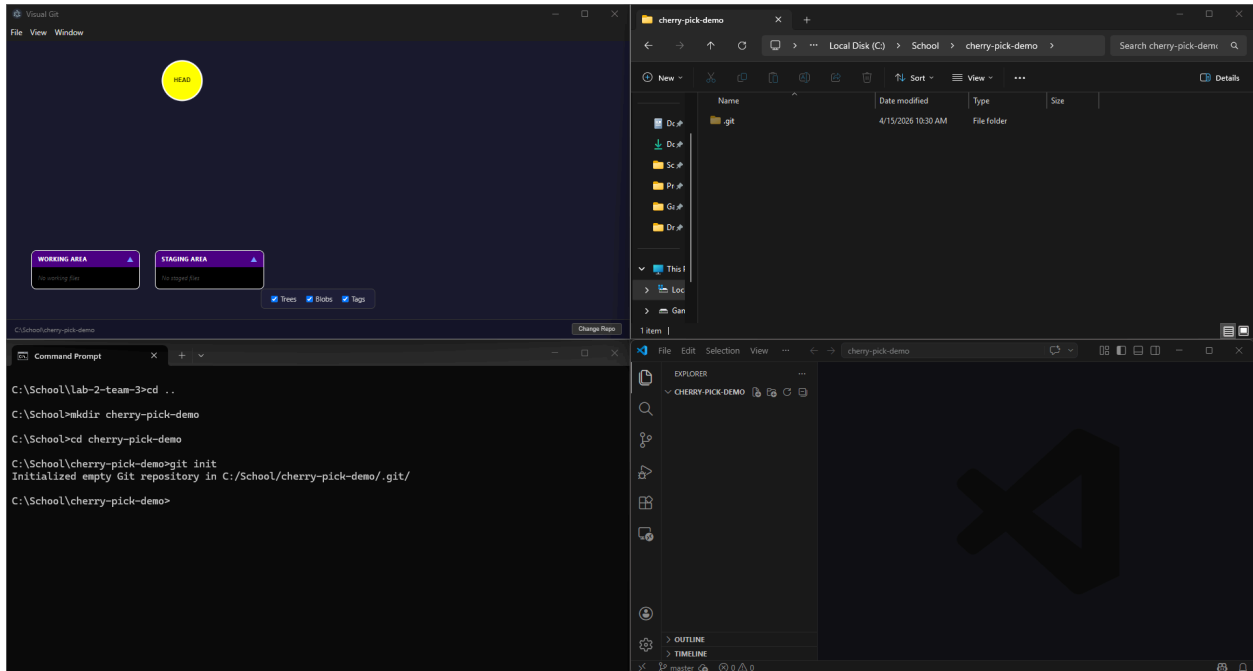
```
git cherry-pick <hash>
<resolve conflict>
```

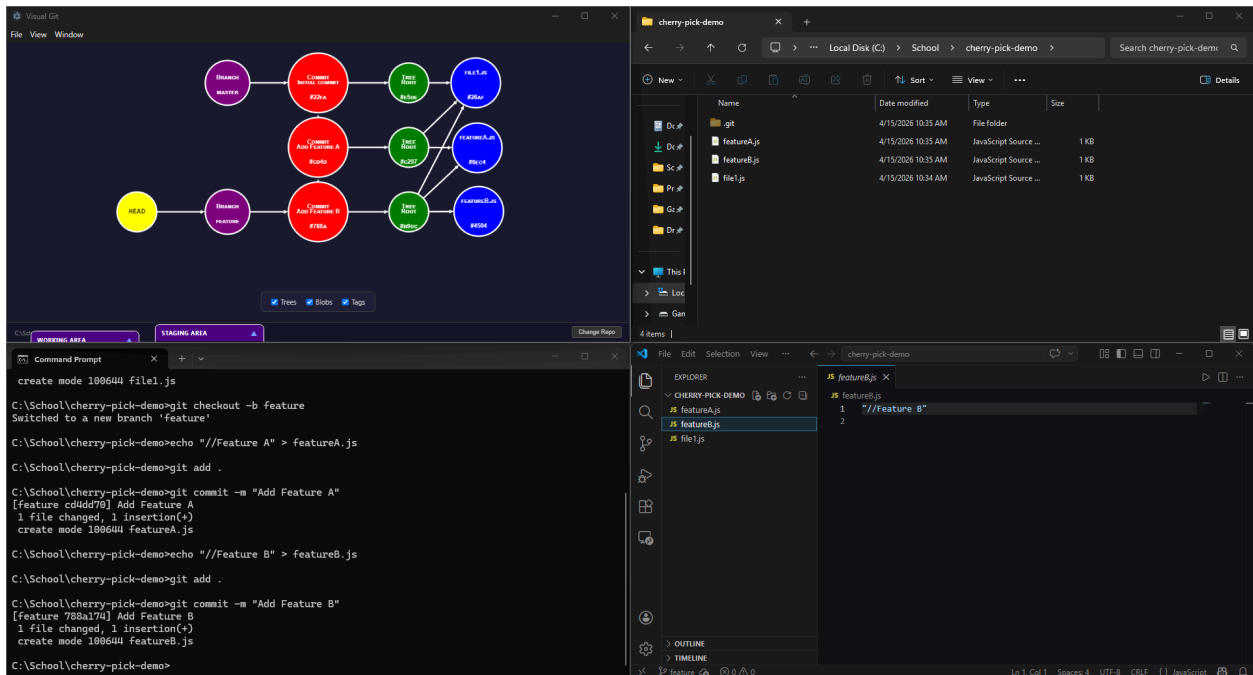
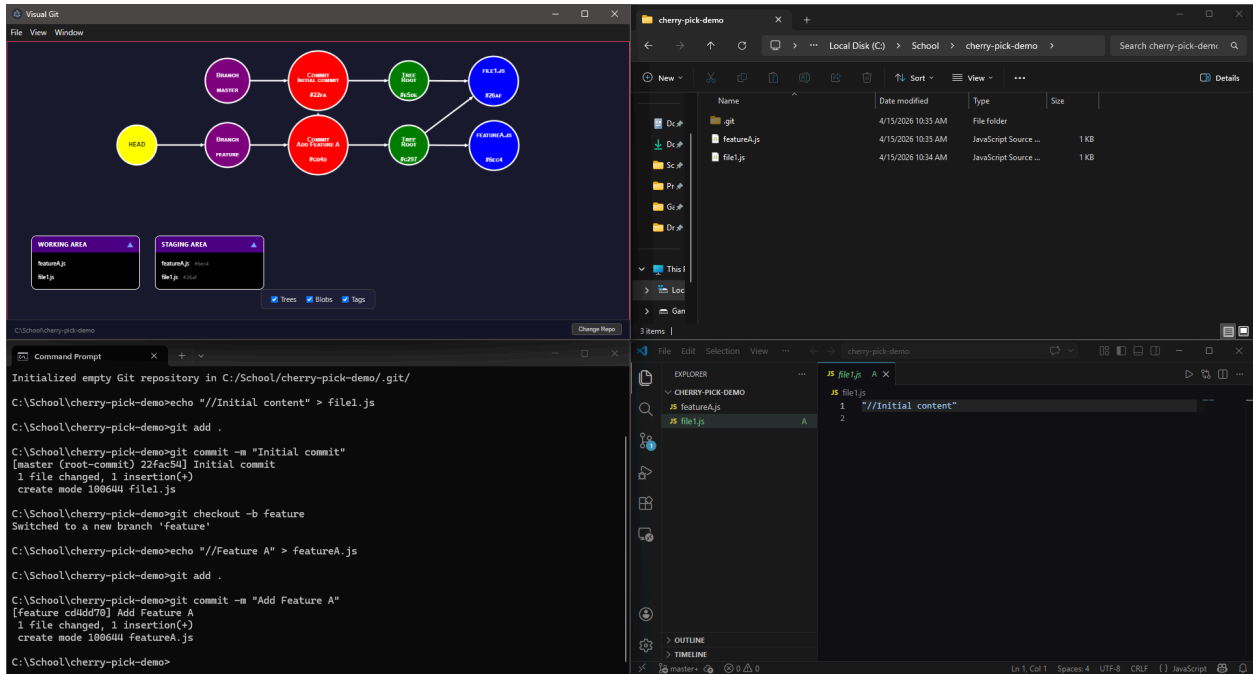
```
git add file1.js
git cherry-pick --continue
```

cherry-pick. I restarted the cherry-pick and resolved it correctly, but VisualGit kept a floating node representing the abandoned attempt. Earlier in the repository's history, I had also pulled in a README file from the remote CS561 repository using the `--allow-unrelated-histories` flag. This created an extra merge commit that appears as a separate branch in VisualGit. These artifacts do not affect the correctness of the cherry-pick scenario; they simply reflect the real sequence of actions I performed.

In the end, the cherry-pick completed successfully, and the final commit history shows the initial commit, the feature branch commits, the conflicting edit on master, and the resolved cherry-pick commit. Even with the extra nodes from earlier mistakes, the repository clearly demonstrates how Git's linked commit structure works, how branches diverge, and how a merge conflict is resolved during a cherry-pick. This completes the assignment requirements, and the VisualGit screenshots illustrate each step of the process.

See visual git below:





Visual Studio interface showing a Git repository named 'cherry-pick-demo'. The left pane displays the Git commit history graph, showing branches (HEAD, master, feature) and commits (42c0b9, 78ba17d, 189644). The right pane shows the file explorer with files: .git, featureA.js, featureB.js, featureC.js, and file1.js. The bottom pane shows the command prompt with the following commands and output:

```
C:\School\cherry-pick-demo>echo "//Feature B" > featureB.js
C:\School\cherry-pick-demo>git add .
C:\School\cherry-pick-demo>git commit -m "Add Feature B"
[feature 78ba17d] Add Feature B
2 file changed, 2 insertion(+)
create mode 189644 featureB.js
C:\School\cherry-pick-demo>echo "//Feature C" > featureC.js
C:\School\cherry-pick-demo>echo "//Initial content" > file1.js
C:\School\cherry-pick-demo>echo "//Feature C edit on feature branch" >> file1.js
C:\School\cherry-pick-demo>git add .
C:\School\cherry-pick-demo>git commit -m "Add Feature C and modify file1.js"
[feature 42c0b9] Add Feature C and modify file1.js
2 files changed, 2 insertion(+)
create mode 189644 featureC.js
C:\School\cherry-pick-demo>
```

Visual Studio interface showing the same 'cherry-pick-demo' repository. The left pane shows the Git commit history graph. The right pane shows the file explorer with files: .git and file1.js. The bottom pane shows the command prompt with the following commands and output:

```
C:\School\cherry-pick-demo>echo "//Initial content" > file1.js
C:\School\cherry-pick-demo>echo "//Feature C edit on feature branch" >> file1.js
C:\School\cherry-pick-demo>git add .
C:\School\cherry-pick-demo>git commit -m "Add Feature C and modify file1.js"
[feature 42c0b9] Add Feature C and modify file1.js
2 files changed, 2 insertion(+)
create mode 189644 featureC.js
C:\School\cherry-pick-demo>git checkout main
error: pathspec 'main' did not match any file(s) known to git
C:\School\cherry-pick-demo>git checkout master
Switched to branch 'master'
C:\School\cherry-pick-demo>git add .
C:\School\cherry-pick-demo>git commit -m "Main branch edit to file1.js"
[master bd43cd4] Main branch edit to file1.js
1 file changed, 1 insertion(+), 1 deletion(-)
C:\School\cherry-pick-demo>
```

Visual Studio interface showing a Git cherry-pick operation. The top-left pane displays a Git commit graph. The top-right pane shows the file explorer for the 'cherry-pick-demo' repository, listing files like .git, featureC.js, and file1.js. The bottom-left pane shows the Command Prompt with the following commands and output:

```
C:\School\cherry-pick-demo>git commit -m "Main branch edit to file1.js"
[master bd43c4f] Main branch edit to file1.js
1 file changed, 1 insertion(+), 1 deletion(-)

C:\School\cherry-pick-demo>git log feature --oneline
42cc0b9 (feature) Add Feature C and modify file1.js
788a17a Add Feature B
cd4d970 Add Feature A
22fac54 Initial commit

C:\School\cherry-pick-demo>git cherry-pick 42cc0b9
Auto-merging file1.js
CONFLICT (content): Merge conflict in file1.js
error: could not apply 42cc0b9... Add Feature C and modify file1.js
hint: After resolving the conflicts, mark them with
hint: 'git add/rm <pathspec>', then run
hint: 'git cherry-pick --continue'.
hint: You can instead skip this commit with 'git cherry-pick --skip'.
hint: To abort and get back to the state before 'git cherry-pick',
hint: run 'git cherry-pick --abort'.
hint: Disable this message with 'git config advice.mergeConflict false'

C:\School\cherry-pick-demo>
```

The bottom-right pane shows the diff view for file1.js, highlighting the conflict between the HEAD (Main branch edit) and the incoming change (Add Feature C and modify file1.js).

Visual Studio Code interface showing a Git repository named 'cherry-pick-demo'. The left sidebar displays the file explorer with 'featureC.js' and 'file1.js'. The main editor shows the 'file1.js' file with a commit message: '//Main branch edit'. The bottom terminal window shows the following commands and output:

```
C:\School\cherry-pick-demo>git commit -m "Main branch edit to file1.js"
[master bd43c4f] Main branch edit to file1.js
1 file changed, 1 insertion(+), 1 deletion(-)

C:\School\cherry-pick-demo>git log --oneline
42cc0b9 (feature) Add Feature C and modify file1.js
788a17a Add Feature B
cd4d070 Add Feature A
22fac54 Initial commit

C:\School\cherry-pick-demo>git cherry-pick 42cc0b9
Auto-merging file1.js
CONFLICT (content): Merge conflict in file1.js
error: could not apply 42cc0b9... Add Feature C and modify file1.js
hint: After resolving the conflicts, mark them with
hint: 'git add/rm <pathspec>', then run
hint: 'git cherry-pick --continue'.
hint: You can instead skip this commit with 'git cherry-pick --skip'.
hint: To abort and get back to the state before 'git cherry-pick',
hint: run 'git cherry-pick --abort'.
hint: Disable this message with 'git config advice.mergeConflict false'

C:\School\cherry-pick-demo>
```

Visual Studio Code interface showing the same 'cherry-pick-demo' repository. The left sidebar displays the file explorer with 'featureC.js' and 'file1.js'. The main editor shows the 'file1.js' file with a commit message: '//Initial content'. The bottom terminal window shows the following commands and output:

```
CONFLICT (content): Merge conflict in file1.js
error: could not apply 42cc0b9... Add Feature C and modify file1.js
hint: After resolving the conflicts, mark them with
hint: 'git add/rm <pathspec>', then run
hint: 'git cherry-pick --continue'.
hint: You can instead skip this commit with 'git cherry-pick --skip'.
hint: To abort and get back to the state before 'git cherry-pick',
hint: run 'git cherry-pick --abort'.
hint: Disable this message with 'git config advice.mergeConflict false'

C:\School\cherry-pick-demo>git cherry-pick 42cc0b9
error: Cherry-picking is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: cherry-pick failed

C:\School\cherry-pick-demo>git add file1.js

C:\School\cherry-pick-demo>git cherry-pick --continue
[master 9b0db1b] merged the conflicts Add Feature C and modify file1.js
Date: Wed Apr 15 10:38:59 2026 -0700
1 file changed, 3 insertions(+), 1 deletion(-)

C:\School\cherry-pick-demo>
```